

Hopf Bifurcation and an Application in Hamiltonian Mechanics

Special Mathematics Lectures – Differential equations and dynamical systems

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Most of the discussion in this report is inspired by Section 11.8, Chapter 11 of [1]. This report serves as a brief introduction to Hopf bifurcation. For more details, see [2, Chapter 4].

1 Hopf Bifurcation

Example 1 (Supercritical Hopf Bifurcation). Consider the system

$$\begin{aligned}\dot{x} &= -y + x(r - x^2 - y^2), \\ \dot{y} &= x + y(r - x^2 - y^2),\end{aligned}\tag{1}$$

where $r \in \mathbb{R}$ is a parameter. This can be written as $(\dot{x}, \dot{y}) = f_r(x, y)$ where the function $f_r : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by

$$f_r(x, y) = \begin{pmatrix} x(r - x^2 - y^2) - y \\ y(r - x^2 - y^2) + x \end{pmatrix}.$$

There is an equilibrium at the origin $(x, y)^t = (0, 0)^t$ for all values of r . The Jacobian matrix at the origin is

$$A(r) = \mathcal{J}_{f_r}(0, 0) = \begin{pmatrix} r - 3x^2 - y^2 & -2xy - 1 \\ -2xy + 1 & r - x^2 - 3y^2 \end{pmatrix} \Big|_{(x, y)^t = (0, 0)^t} = \begin{pmatrix} r & -1 \\ 1 & r \end{pmatrix}.$$

The eigenvalues of $A(r)$ are solutions of the characteristic polynomial

$$\det(A(r) - \lambda \mathbf{1}_2) = \det \begin{pmatrix} r - \lambda & -1 \\ 1 & r - \lambda \end{pmatrix} = (r - \lambda)^2 + 1 = 0,$$

which gives two eigenvalues λ and $\bar{\lambda}$ where λ is given by

$$\lambda = r + i.$$

Observe that $r = \operatorname{Re}(\lambda) = \operatorname{Re}(\bar{\lambda})$. By Theorem 4.3.5 in the lecture notes [3], the equilibrium at the origin is exponentially stable when $r < 0$ and exponentially unstable when $r > 0$.

Furthermore, in the case $r > 0$, there exists a 2π -periodic orbit $\Gamma = \gamma((\sqrt{r}, 0)^t)$ which elements are given by

$$\begin{aligned}x_\Gamma(t) &= \sqrt{r} \cos(t), \\ y_\Gamma(t) &= \sqrt{r} \sin(t).\end{aligned}$$

We can verify this fact by substituting this solution into (1).

For the next step, consider the hypersurface $\Sigma = \{(x, y)^t \in \mathbb{R}^2 \mid y = 0\}$, which is orthogonal to Γ at $(\sqrt{r}, 0)$. To simplify the notation, we identify Σ with $\mathbb{R}$. Then, to study the stability of Γ , we compute the Poincaré map $\mathcal{P} : (\sqrt{r} - \delta, \sqrt{r} + \delta) \rightarrow \mathbb{R}$ for some small enough $\delta > 0$

so that $\sqrt{r} - \delta > 0$. In order to obtain $\mathcal{P}$, we solve the system (1) with initial condition $(x(0), y(0))^t = (x_0, 0)^t$ for $x_0 \in (\sqrt{r} - \delta, \sqrt{r} + \delta)$. This can be done by switching to polar coordinates (ρ, θ) where

$$\begin{aligned} x(t) &= \rho(t) \cos(\theta(t)), \\ y(t) &= \rho(t) \sin(\theta(t)). \end{aligned} \tag{2}$$

We can derive a system of differential equations for ρ and θ as follows. First, using (1), we compute

$$\dot{\rho} = \frac{d}{dt} \sqrt{x^2 + y^2} = \frac{x\dot{x} + y\dot{y}}{\sqrt{x^2 + y^2}} = \frac{-xy + x^2(r - \rho^2) + xy + y^2(r - \rho^2)}{\rho} = \rho(r - \rho^2),$$

where we assume that $\rho \neq 0$. The assumption $\rho \neq 0$ can be justified by the fact that the origin is an unstable fixed point for $r > 0$. Next, we have

$$x\dot{y} - \dot{x}y = \rho \cos(\theta)(\dot{\rho} \sin(\theta) + \rho \cos(\theta)\dot{\theta}) - (\dot{\rho} \cos(\theta) - \rho \sin(\theta)\dot{\theta})\rho \sin(\theta) = \rho^2 \dot{\theta}$$

$$\Leftrightarrow \quad \dot{\theta} = \frac{x\dot{y} - \dot{x}y}{\rho^2}.$$

Then, using (1) again, we obtain

$$\dot{\theta} = \frac{x(x + y(r - \rho^2)) - (-y + x(r - \rho^2))y}{\rho^2} = \frac{x^2 + y^2}{\rho^2} = 1.$$

Therefore, we have derived the system

$$\begin{aligned} \dot{\rho} &= \rho(r - \rho^2), \\ \dot{\theta} &= 1. \end{aligned} \tag{3}$$

The initial condition $(x(0), y(0)) = (x_0, 0)$ can be translated into an initial condition

$$(\rho(0), \theta(0))^t = (x_0, 0)^t.$$

The second equation in (3) has a simple solution $\theta(t) = t$. Observe that $\rho(t) = \sqrt{r}$ is an equilibrium solution of the first equation in (3), which corresponds to the periodic orbit Γ . Now, assuming that $\rho < \sqrt{r}$, we can integrate that equation as follows:

$$\int \frac{1}{\rho(r - \rho^2)} d\rho = \int dt + C.$$

for some constant C . The left-hand side can be computed using variable substitution $\xi = \frac{1}{\rho^2}$:

$$\int \frac{1}{\rho(r - \rho^2)} d\rho = -\frac{1}{2} \int \frac{1}{r\xi - 1} d\xi = -\frac{1}{2r} \ln(r\xi - 1) = -\frac{1}{2r} \ln\left(\frac{r}{\rho^2} - 1\right).$$

Therefore, we have

$$-\frac{1}{2r} \ln\left(\frac{r}{\rho^2} - 1\right) = t + C \quad \Leftrightarrow \quad \rho(t) = \frac{\sqrt{r}}{\sqrt{1 + Ae^{-2rt}}},$$

where $A = e^{-2rC}$. The constant A can be determined using the initial condition $\rho(0) = x_0$. This gives

$$\rho(t) = \frac{x_0 \sqrt{r}}{\sqrt{x_0^2 + (r - x_0^2)e^{-2rt}}}.$$

We can also obtain the same solution by assuming that $\rho > \sqrt{r}$. Hence, the Poincaré map evaluated at x_0 is given by

$$\mathcal{P}(x_0) = \rho(2\pi) = \frac{x_0 \sqrt{r}}{\sqrt{x_0^2 + (r - x_0^2)e^{-4\pi r}}}.$$

Finally, we compute the derivative of $\mathcal{P}$ at the fixed point $x_0 = \sqrt{r}$:

$$\begin{aligned}\mathcal{P}'(\sqrt{r}) &= \left. \frac{d\mathcal{P}}{dx_0} \right|_{x_0=\sqrt{r}} = \left[\frac{\sqrt{r}}{\sqrt{x_0^2 + (r - x_0^2)e^{-4\pi r}}} - \frac{\sqrt{r}(x_0^2 - x_0^2 e^{-4\pi r})}{(x_0^2 + (r - x_0^2)e^{-4\pi r})^{3/2}} \right] \bigg|_{x_0=\sqrt{r}} \\ &= \frac{r\sqrt{r}e^{-4\pi r}}{(x_0^2 + (r - x_0^2)e^{-4\pi r})^{3/2}} \bigg|_{x_0=\sqrt{r}} = e^{-4\pi r}.\end{aligned}$$

According to Theorem 6.3.5 in the lecture notes [3], the periodic orbit Γ is asymptotically stable when $r > 0$ since $|\mathcal{P}'(\sqrt{r})| < 1$. Hence, as the parameter r crosses the critical value $r = 0$ from negative to positive values, the fixed point at the origin loses its stability and a stable periodic orbit Γ emerges from the fixed point. This transition from $r < 0$ to $r > 0$ is continuous as the radius of Γ increases from 0 when r increases from 0. This phenomenon is known as a *supercritical Hopf bifurcation*.

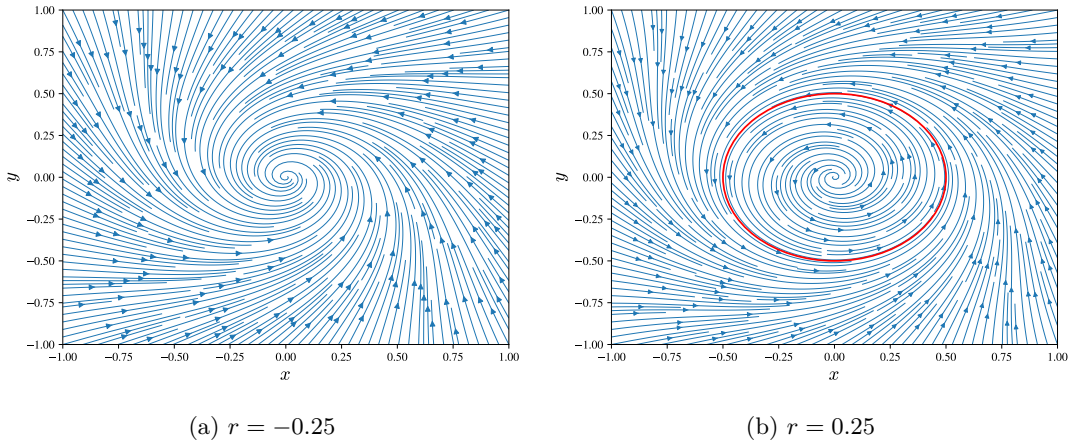


Figure 1: Integral curves of the system (1) are shown for $r < 0$ and $r > 0$. When $r < 0$, the fixed point at the origin is attracting. When $r > 0$, a stable periodic orbit (red circle) emerges surrounding the unstable fixed point at the origin.

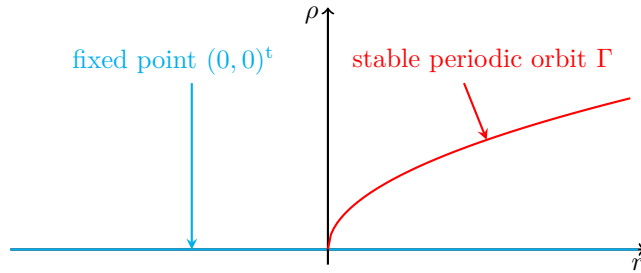


Figure 2: A diagram illustrates the supercritical Hopf bifurcation in the system (1).

In Example 1, we computed the Poincaré map explicitly and examined its derivative at the periodic orbit to determine its stability. However, in general, it is not always possible to obtain an explicit formula for the Poincaré map. The following theorem provides a convenient method to compute the derivative of the Poincaré map at a periodic orbit without having to find its explicit expression [2, Theorem 2, pp. 216].

Theorem 1. Consider $M \subset \mathbb{R}^2$. Let $(\mathbb{R}, M, \phi)$ be the dynamical system defined by the autonomous first order differential system $\dot{x}(t) = f(x(t))$ with $f \in C^1(M, \mathbb{R}^2)$. Assume that $\Gamma = \gamma(x_0)$ is a T -periodic orbit, and let $\mathcal{P} : B_\delta(x_0) \cap \Sigma \rightarrow \Sigma$ be the associated Poincaré map at x_0 where Σ is a straight line orthogonal to Γ at x_0 . By identifying Σ with $\mathbb{R}$ and x_0 with

$0 \in \mathbb{R}$, we can consider $\mathcal{P} : (-\delta, \delta) \rightarrow \mathbb{R}$. Then, the derivative of $\mathcal{P}$ at x_0 is given by

$$\mathcal{P}'(0) = \exp \left(\int_0^T [\operatorname{div} f](x_\Gamma(t)) dt \right),$$

where $x_\Gamma(t)$ is the periodic solution corresponding to the periodic orbit Γ .

We illustrate the usage of this theorem in the following example.

Example 2 (Subcritical Hopf Bifurcation with hysteresis). Consider the system

$$\begin{aligned} \dot{x} &= -y + x [r + 2(x^2 + y^2) - (x^2 + y^2)^2], \\ \dot{y} &= x + y [r + 2(x^2 + y^2) - (x^2 + y^2)^2], \end{aligned} \quad (4)$$

which can be written as $(\dot{x}, \dot{y}) = f_r(x, y)$ where the function $f_r : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by

$$f_r(x, y) = \begin{pmatrix} -y + x [r + 2(x^2 + y^2) - (x^2 + y^2)^2] \\ x + y [r + 2(x^2 + y^2) - (x^2 + y^2)^2] \end{pmatrix}.$$

This system also has an equilibrium at the origin $(x, y)^t = (0, 0)^t$ for all values of r . The Jacobian matrix at the origin is

$$\begin{aligned} A(r) &= \mathcal{J}_{f_r}(0, 0) \\ &= \begin{pmatrix} r + 2(3x^2 + y^2) - (5x^4 + y^4 + 6x^2y^2) & -1 + 4xy - 4xy(x^2 + y^2) \\ 1 + 4xy - 4xy(x^2 + y^2) & r + 2(x^2 + 3y^2) - (x^4 + 5y^4 + 6x^2y^2) \end{pmatrix} \Big|_{\substack{x=0 \\ y=0}} \\ &= \begin{pmatrix} r & -1 \\ 1 & r \end{pmatrix}. \end{aligned}$$

The eigenvalues of $A(r)$ are the same as in Example 1, which implies that this fixed point goes from being exponentially stable to exponentially unstable as r crosses the critical value $r = 0$ from negative to positive values.

This system can also be transformed into polar coordinates (ρ, θ) by using (2). Following the same procedure as in Example 1, we have

$$\begin{aligned} \dot{\rho} &= \frac{x\dot{x} + y\dot{y}}{\sqrt{x^2 + y^2}} = \frac{(x^2 + y^2) [r + 2(x^2 + y^2) - (x^2 + y^2)^2]}{\rho} = \rho(r + 2\rho^2 - \rho^4), \\ \dot{\theta} &= \frac{x\dot{y} - \dot{x}y}{\rho^2} = \frac{x^2 + y^2}{\rho^2} = 1. \end{aligned}$$

Hence, we obtain the system

$$\begin{aligned} \dot{\rho} &= \rho(r + 2\rho^2 - \rho^4) \\ \dot{\theta} &= 1 \end{aligned} \quad (5)$$

An equilibrium solution $\rho(t) = \rho_0$ of the first equation of (5) is obtained if ρ_0 satisfies the equation

$$\rho_0(r + 2\rho_0^2 - \rho_0^4) = 0.$$

For $r > 0$, the solution $\rho_0 = 0$ corresponds to the unstable fixed point at the origin. Hence, we are interested in a non-zero solution ρ_0 which satisfies

$$\rho_0^4 - 2\rho_0^2 - r = 0,$$

which has only one solution given by

$$\rho_0 = \rho_+(r) = \sqrt{1 + \sqrt{1 + r}}$$

for $r > 0$. In fact, for r such that $-1 < r < 0$, while the solution $\rho_0 = \rho_+(r)$ is still valid, there exists another solution given by

$$\rho_0 = \rho_-(r) = \sqrt{1 - \sqrt{1 + r}}.$$

These two solutions $\rho_{\pm}(r)$ correspond to two 2π -periodic orbits $\Gamma_{\pm}$ surrounding the fixed point at the origin when $-1 < r < 0$. To study the stability of these periodic orbits, we choose the hypersurface $\Sigma = \{(x, y)^t \in \mathbb{R}^2 \mid y = 0\} \cong \mathbb{R}$ and compute the Poincaré map $\mathcal{P}$ for each periodic orbit. First, following the same process as in Example 1, we attempt to solve the system (4) with initial condition $(\rho(0), \theta(0)) = (x_0, 0)$ for $x_0 > 0$. From (5), we obtain $\theta(t) = t$ and

$$\int_{x_0}^{\rho(t)} \frac{1}{\rho(r + 2\rho^2 - \rho^4)} d\rho = t.$$

Then, the Poincaré map $\mathcal{P}$ defined by $\mathcal{P}(x_0) = \rho(2\pi)$ can be expressed implicitly as

$$\int_x^{\mathcal{P}(x)} \frac{1}{\rho(r + 2\rho^2 - \rho^4)} d\rho = 2\pi,$$

where we change x_0 into x for simplicity. We can evaluate the integral on the left-hand side:

$$\int \frac{1}{\rho(r + 2\rho^2 - \rho^4)} d\rho = -\frac{1}{4r} \ln \left| \frac{r + 2\rho^2 - \rho^4}{\rho^4} \right| + \frac{1}{4r\sqrt{1+r}} \ln \left| \frac{r/\rho^2 + 1 - \sqrt{1+r}}{r/\rho^2 + 1 + \sqrt{1+r}} \right|.$$

However, we can see that it is difficult to invert this expression to obtain an explicit formula for $\mathcal{P}(x)$. Instead, we can compute the derivative of $\mathcal{P}$ at each orbit $\Gamma_{\pm}$ using Theorem 1. First, we compute

$$\begin{aligned} [\operatorname{div} f_r](x, y) &= \left. \frac{\partial f_r}{\partial x} \right|_{(x, y)^t} + \left. \frac{\partial f_r}{\partial y} \right|_{(x, y)^t} \\ &= r + 2(3x^2 + y^2) - (5x^4 + y^4 + 6x^2y^2) + r + 2(x^2 + 3y^2) - (x^4 + 5y^4 + 6x^2y^2) \\ &= 2r + 8(x^2 + y^2) - 6(x^2 + y^2)^2 \\ &= 2r + 8\rho^2 - 6\rho^4. \end{aligned}$$

For $\rho = \rho_{\pm}(r) \equiv \rho_{\pm}$, we have

$$\begin{aligned} [\operatorname{div} f_r](x_{\pm}, y_{\pm}) &= 6(r + 2\rho_{\pm}^2 - \rho_{\pm}^4) - 4(r + \rho_{\pm}^2) = 0 - 4(r + 1 \pm \sqrt{1+r}) \\ &= -4\sqrt{1+r} (\sqrt{1+r} \pm 1). \end{aligned}$$

Therefore, using Theorem 1, we obtain

$$\mathcal{P}'(\rho_{\pm}) = \exp \left(\int_0^{2\pi} [\operatorname{div} f_r](x_{\pm}(t), y_{\pm}(t)) dt \right) = \exp [-8\pi\sqrt{1+r} (\sqrt{1+r} \pm 1)].$$

By Theorem 6.3.5 in the lecture note [3], the fact that $\mathcal{P}'(\rho_+) < 1$ for all $r > -1$ implies that the periodic orbit Γ_+ is always asymptotically stable. On the other hand, since $\mathcal{P}'(\rho_-) > 1$ for all $-1 < r < 0$, the periodic orbit Γ_- is always unstable.

The behavior of the system (4) is more complicated than the system in Example 1. As the parameter r crosses the critical value $r = 0$ from negative to positive values, the unstable orbit Γ_- shrinks into the origin making the fixed point repulsive while the stable orbit Γ_+ keeps increasing in size. Hence, after the bifurcation point $r = 0$, the fixed point suddenly becomes unstable and all orbits are attracted to Γ_+ . This phenomenon is known as a *subcritical Hopf bifurcation*. In addition, when $-1 < r < 0$, there are an attracting fixed point and an attracting periodic orbit coexisting with each other. If we change the parameter r in this range and then return it to its original value, the system may end up in a different state than the initial one. This phenomenon is known as *hysteresis*. This can be explained by supposing that the system is attracted at first to the fixed point. As r is increased past $r = 0$, the fixed point becomes

repulsive and the system is then attracted to the stable periodic orbit Γ_+ . If r is then decreased back to its original value, the system remains attracted to Γ_+ since the unstable periodic orbit Γ_- prevents it from approaching the fixed point again.

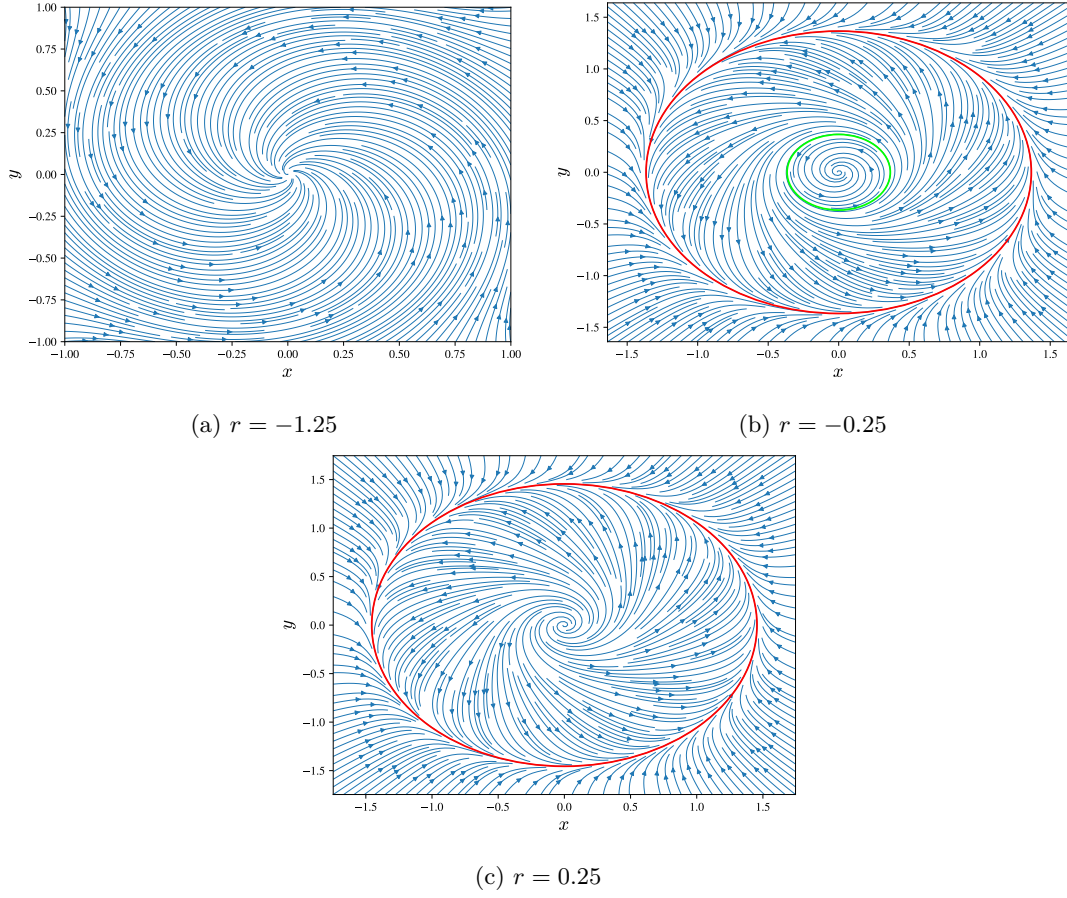


Figure 3: Integral curves of the system (4) are shown for $r < -1$, $-1 < r < 0$, and $r > 0$.

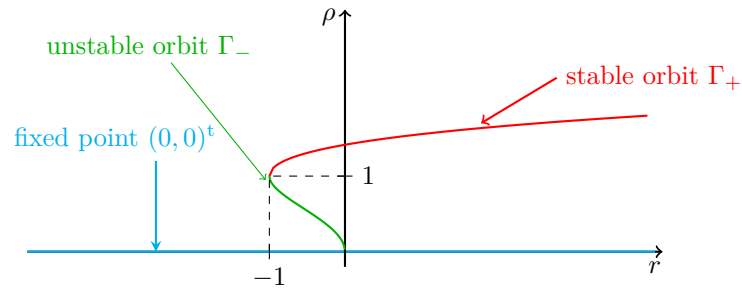


Figure 4: A diagram illustrates the subcritical Hopf bifurcation with hysteresis in the system (4).

In last two examples, we observe that a periodic orbit emerges from the fixed point at the bifurcation point and its size changes as the parameter r varies. However, this is not always the case as shown in the following example.

Example 3. Consider the system

$$\begin{aligned}\dot{x} &= y + rx, \\ \dot{y} &= -x + ry,\end{aligned}\tag{6}$$

with the corresponding vector field $f_r : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$f_r(x, y) = \begin{pmatrix} y + rx \\ -x + ry \end{pmatrix}.$$

The Jacobian matrix at the fixed point $(0, 0)^t$ is

$$A(r) = \mathcal{J}_{f_r}(0, 0) = \begin{pmatrix} r & 1 \\ -1 & r \end{pmatrix},$$

which has the same eigenvalues as in Example 1 and Example 2. However, in this case, we have no periodic orbits surrounding the fixed point for any nonzero value of r . In fact, the system (6) can be solved explicitly since it is linear. One trick to solve it is to consider the complex variable $z(t) = x(t) + iy(t)$. Then, the system (6) can be written as a single complex differential equation

$$\dot{z} = \dot{x} + i\dot{y} = -i(x + iy) + r(x + iy) = (i + r)z.$$

This equation can be integrated directly to obtain

$$z(t) = z(0)e^{(i+r)t} = z(0)e^{rt}(\cos t + i \sin t).$$

Therefore, the solution of the system (6) with initial condition $(x(0), y(0)) = (\rho, 0)$ for $\rho \geq 0$ is given by

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^{rt} \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \begin{pmatrix} \rho \\ 0 \end{pmatrix} = e^{rt} \begin{pmatrix} \rho \cos t \\ \rho \sin t \end{pmatrix}.$$

When $r \neq 0$, this solution is either spiraling outwards or inwards depending on the sign of r , which shows that there is no periodic orbit in this case. When $r = 0$, it is periodic with period 2π since it corresponds to a uniform circular motion around the origin. Furthermore, there is a family of periodic orbits Γ_ρ labeled by their radii $\rho \neq 0$. As ρ approaches 0, these periodic orbits Γ_ρ converge to the fixed point at the origin, which is also a Hopf bifurcation.

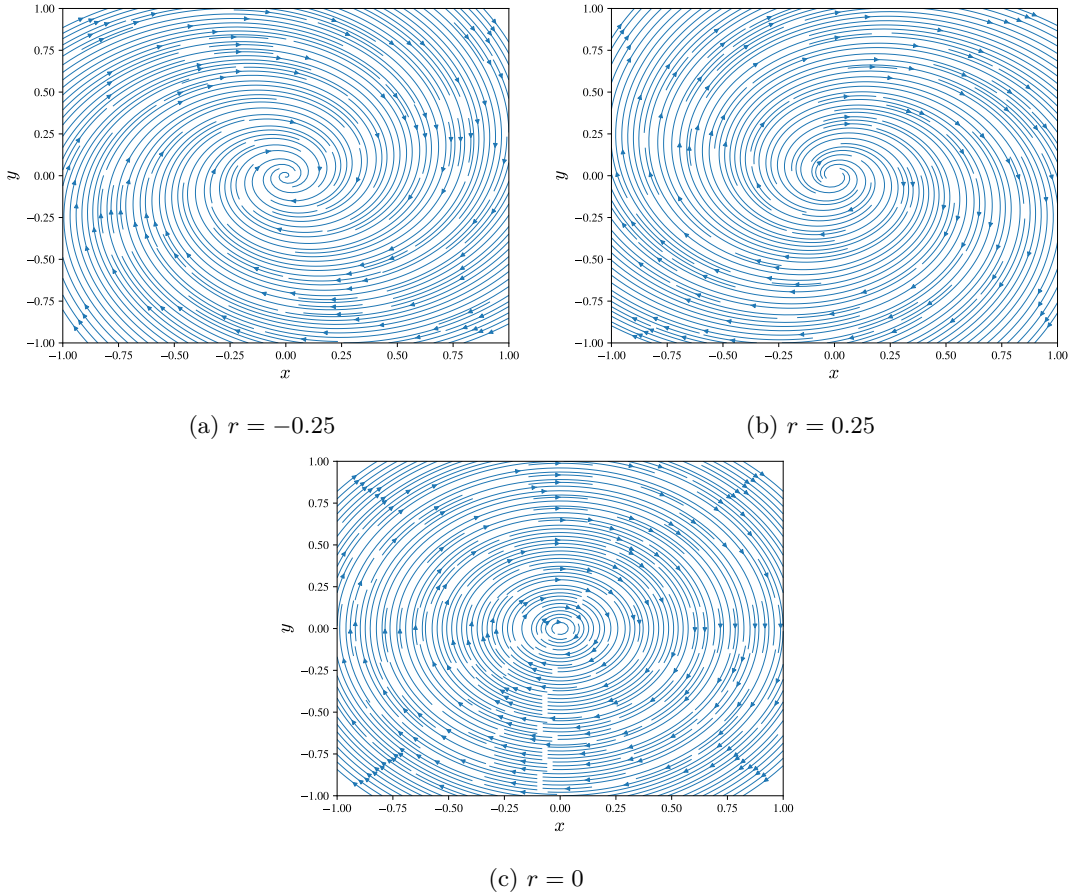


Figure 5: Integral curves of the system (6) are shown for $r < 0$, $r > 0$, and $r = 0$.

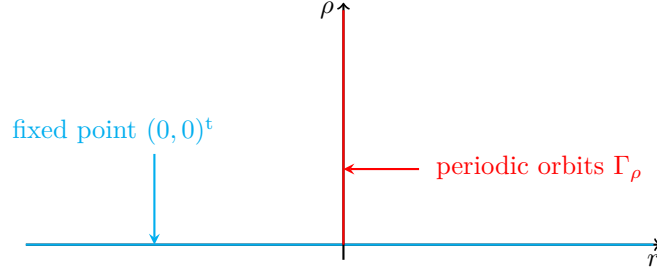


Figure 6: A diagram illustrates the Hopf bifurcation in the system (6).

To summarize the discussion above, we introduce the following definition.

Definition 2. Assume that $x = x_0$ is an equilibrium of a n -dimensional differential system $\dot{x}(t) = f_r(x(t))$ at $r = r_0$. We say that a *path of periodic orbits bifurcates from* (r_0, x_0) if there is a continuous path of periodic orbits that converge to the equilibrium at $r = r_0$ as seen from the space of all values of $(r, x) \in \mathbb{R}^{n+1}$.

As seen from Example 1, a stable periodic orbit Γ bifurcates from the equilibrium $(0, 0)^t$ while in Example 2, an unstable periodic orbit Γ_- bifurcates from the same equilibrium. Meanwhile, in Example 3, a family of periodic orbits Γ_ρ bifurcates from the origin entirely on the plane $r = 0$. In general, when a path of periodic orbits bifurcates from an equilibrium as the parameter r crosses a critical value r_0 , we say that an *Andronov-Hopf bifurcation*, or a *Hopf bifurcation* in short, occurs at (r_0, x_0) . The Hopf bifurcation of a system is supercritical if an attracting periodic orbit bifurcates from the equilibrium, and subcritical if a repulsive periodic orbit bifurcates from the fixed point. Therefore, Example 1 is supercritical and Example 2 is subcritical while Example 3 is neither supercritical nor subcritical. The following theorem provides sufficient conditions for the occurrence of a Hopf bifurcation [1, Theorem 11.21][2, Theorem 2, pp. 353].

Theorem 3 (Andronov-Hopf Bifurcation Theorem). *Let $\dot{x}(t) = f_r(x(t))$ be a one-parameter family of differential equations in $\mathbb{R}^n$ with an equilibrium at $x = x_0(r)$ for each value of the parameter $r \in \mathbb{R}$. Assume that the Jacobian matrix $A(r) = \mathcal{J}_{f_r}(x_0(r))$ has a pair of complex conjugate eigenvalues $a(r) \pm ib(r)$ that crosses the imaginary axis at $r = r_0$ with a non-zero velocity, i.e. $a(r_0) = 0$, $b := b(r_0) \neq 0$, $a'(r_0) \neq 0$, and $a(r_1)a(r_2) < 0$ for r_1, r_2 close to r_0 such that $r_1 < r_0 < r_2$. Assume further that no other eigenvalue of $A(r_0)$ is an integer multiple of ib . Then a path of periodic orbits bifurcates from $(r, x) = (r_0, x_0(r_0))$. The periods of these orbits approach $2\pi/b$ as they approach the equilibrium.*

Example 4 (Lorenz System). Consider the Lorenz system defined by the differential equations

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= rx - y - xz, \\ \dot{z} &= xy - bz,\end{aligned}\tag{7}$$

where σ , r , and b are positive parameters. The system has an equilibrium at the origin $(x, y, z) = (0, 0, 0)$ for all values of r . As shown in the lecture note [3], this fixed point is exponentially stable when $0 < r < 1$. The eigenvalues of the Jacobian matrix at the origin are given by $\lambda_1 = -b$ and

$$\lambda_{\pm} = \frac{-(\sigma + 1) \pm \sqrt{(\sigma + 1)^2 - 4\sigma(1 - r)}}{2}.$$

All of these eigenvalues are negative for $0 < r < 1$. However, when $r > 1$, the eigenvalue λ_+ become positive, making the fixed point no more stable. In addition, for $r > 1$, there are two new fixed points at

$$c_{\pm} = \left(\pm\sqrt{b(r-1)}, \pm\sqrt{b(r-1)}, r-1 \right)^t.$$

The Jacobian matrix at these fixed points is given by

$$A_{\pm}(r) = \begin{pmatrix} -\sigma & \sigma & 0 \\ r-z & -1 & -x \\ y & x & -b \end{pmatrix} \Big|_{(x,y,z)^t = c_{\pm}} = \begin{pmatrix} -\sigma & \sigma & 0 \\ 1 & -1 & \mp \sqrt{b(r-1)} \\ \pm \sqrt{b(r-1)} & \pm \sqrt{b(r-1)} & -b \end{pmatrix}.$$

The characteristic polynomial of $A_{\pm}(r)$ is computed as follows:

$$\begin{aligned} p(\lambda) &= \det(\lambda I - A_{\pm}(r)) \\ &= \begin{vmatrix} \lambda + \sigma & -\sigma & 0 \\ -1 & \lambda + 1 & \pm \sqrt{b(r-1)} \\ \mp \sqrt{b(r-1)} & \mp \sqrt{b(r-1)} & \lambda + b \end{vmatrix} \\ &= (\lambda + \sigma) \begin{vmatrix} \lambda + 1 & \pm \sqrt{b(r-1)} \\ \mp \sqrt{b(r-1)} & \lambda + b \end{vmatrix} + \sigma \begin{vmatrix} -1 & \pm \sqrt{b(r-1)} \\ \mp \sqrt{b(r-1)} & \lambda + b \end{vmatrix} + 0 \\ &= (\lambda + \sigma) [(\lambda + 1)(\lambda + b) + b(r-1)] + \sigma [-(\lambda + b) + b(r-1)] \\ &= \lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0, \end{aligned}$$

where $a_2 = \sigma + b + 1$, $a_1 = b(r + \sigma)$, and $a_0 = 2b\sigma(r - 1)$. For $r > 1$, all coefficients a_0, a_1, a_2 are positive. The polynomial $p(\lambda)$ has a pair of purely imaginary roots if and only if $a_0 = a_1 a_2$. Indeed, if $a_0 = a_1 a_2$, then we have

$$p(\lambda) = \lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_1 a_2 = (\lambda^2 + a_1)(\lambda + a_2),$$

which has roots $\lambda = -a_2$ and $\lambda = \pm i\sqrt{a_1}$. Conversely, if $p(\lambda)$ has a real root $\lambda = a$ and a pair of purely imaginary roots $\lambda = \pm ic$, then

$$p(\lambda) = (\lambda - ic)(\lambda + ic)(\lambda - a) = \lambda^3 - a\lambda^2 + c^2\lambda - ac^2,$$

which shows that $a_0 = -ac^2 = a_1 a_2$. The condition $a_0 = a_1 a_2$ gives us a value of r at which a subcritical Hopf bifurcation occurs:

$$2b\sigma(r - 1) = b(r + \sigma)(\sigma + b + 1) \quad \Leftrightarrow \quad r = \frac{\sigma(\sigma + b + 3)}{\sigma - b - 1} =: r_0.$$

The corresponding pure imaginary eigenvalues are

$$\pm i\sqrt{a_1} = \pm i\sqrt{b(r_0 + \sigma)} = \pm i\sqrt{\frac{2b\sigma(\sigma + 1)}{\sigma - b - 1}}.$$

By [Routh–Hurwitz criterion](#) for third-order polynomials, all roots of $p(\lambda)$ have negative real parts if and only if $a_2 > 0$, $a_0 > 0$, and $a_0 < a_1 a_2$, which holds when $1 < r < r_0$. Therefore, the fixed points $c_{\pm}$ are exponentially stable for $1 < r < r_0$ and unstable for $r > r_0$. To show that the bifurcation is subcritical, we need to perform an appropriate coordinate transformation and put the equations in a certain form, which is outside the scope of this report. For classical values of the parameters $\sigma = 10$ and $b = 8/3$, we have $r_0 = \frac{470}{19} \approx 24.74$ and $\sqrt{a_1} = \sqrt{\frac{1760}{19}} \approx 9.62$. Furthermore, unstable periodic orbits around $c_{\pm}$ start to appear from $r \approx 13.926$ [[1](#), Example 11.25]. These orbits converge to $c_{\pm}$ at $r = r_0$ in the same manner as Γ_- in Example [2](#).

2 An Application in Hamiltonian Mechanics

For the discussion in this section, we only need a weaker version of Theorem [3](#) with weaker assumption and conclusion [[1](#), Remark 11.22].

Theorem 4 (Andronov-Hopf Bifurcation Theorem – Weak Form). *Let $\dot{x}(t) = f_r(x(t))$ be a one-parameter family of differential equations in $\mathbb{R}^n$ satisfying all assumptions of Theorem [3](#) except that there is no constraint on $a'(r_0)$. Then every neighborhood of $(r, x) = (r_0, x_0(r_0))$, no matter how small, contains a periodic orbit. The periods of these orbits approach $2\pi/b$ as they approach the equilibrium.*

Remark. This theorem guarantees the existence of a connected set of periodic orbits emanating from $(r_0, x_0(r_0))$ but it does not guarantee that this set of periodic orbits can be parameterized continuously. Nevertheless, we still say that a Hopf bifurcation occurs at $(r_0, x_0(r_0))$.

Now, let us recall the definition of *autonomous Hamiltonian systems*, or Hamiltonian systems in short, from the report [4]. For an integer $n \geq 1$, consider a function $\mathbb{R}^n \times \mathbb{R}^n \ni (q, p) \mapsto \mathcal{H}(q, p) \in \mathbb{R}$ of class C^2 and the differential system $\dot{x} = f(x)$ where $x(t) = (q(t), p(t))^t$ and $f : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ given by

$$f = \left(\frac{\partial \mathcal{H}}{\partial p_1}, \dots, \frac{\partial \mathcal{H}}{\partial p_n}, -\frac{\partial \mathcal{H}}{\partial q_1}, \dots, -\frac{\partial \mathcal{H}}{\partial q_n} \right)^t = J \nabla \mathcal{H}, \quad (8)$$

where the $2n \times 2n$ matrix J is given by

$$J = \begin{pmatrix} 0 & \mathbb{1}_n \\ -\mathbb{1}_n & 0 \end{pmatrix}.$$

Such a system is called an autonomous *Hamiltonian system* with *Hamiltonian* $\mathcal{H}$. The following lemma shows that the value of the Hamiltonian $\mathcal{H}$ evaluated along each orbit is a constant of motion.

Lemma 5. *Let $\dot{x}(t) = f(x(t))$ be an autonomous differential system in $\mathbb{R}^{2n}$ with $f : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ given by (8) for a function $\mathcal{H} : \mathbb{R}^{2n} \rightarrow \mathbb{R}$ of class C^2 . Then, for every solution $x(t)$ of this system, the function $t \mapsto \mathcal{H}(x(t))$ is a constant function.*

Proof. Using the chain rule and the equation $\dot{x}(t) = f(x(t))$, we have

$$\frac{d}{dt} \mathcal{H}(q(t), p(t)) = \sum_{i=1}^n \frac{\partial \mathcal{H}}{\partial q_i} \dot{q}_i(t) + \sum_{i=1}^n \frac{\partial \mathcal{H}}{\partial p_i} \dot{p}_i(t) = \sum_{i=1}^n \frac{\partial \mathcal{H}}{\partial q_i} \left(\frac{\partial \mathcal{H}}{\partial p_i} \right) + \sum_{i=1}^n \frac{\partial \mathcal{H}}{\partial p_i} \left(-\frac{\partial \mathcal{H}}{\partial q_i} \right) = 0.$$

These calculations can be rewritten conveniently as

$$\frac{d}{dt} \mathcal{H}(x(t)) = \nabla \mathcal{H}(x(t)) \cdot \dot{x}(t) = \nabla \mathcal{H}(x(t)) \cdot J \nabla \mathcal{H}(x(t)) = 0. \quad \square$$

An equilibrium point x_0 is called a *center* when the Jacobian matrix at x_0 has two pure imaginary eigenvalues. The following theorem shows the existence of periodic orbits around a center of a Hamiltonian system [1, Challenge 11].

Theorem 6 (Lyapunov Center Theorem). *Assume that x_0 is a center equilibrium of the Hamiltonian system and that $\pm bi$ are simple eigenvalues of the Jacobian matrix $A_0 = \mathcal{J}_f(x_0)$ at x_0 . Assume further that no other eigenvalue of A_0 is an integer multiple of bi . Then each neighborhood of the center contains periodic orbits, whose periods approach $2\pi/b$ as they approach the center.*

Proof. We embed the Hamiltonian system in a one-parameter family of differential equation $\dot{x}(t) = f_r(x(t))$ where the vector field $f_r : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ is given by

$$f_r = (J + r\mathbb{1}_{2n}) \nabla \mathcal{H}.$$

The original Hamiltonian system occurs at $r = 0$. Since x_0 is an equilibrium of the Hamiltonian system, we have

$$f_0(x_0) = 0 \quad \Leftrightarrow \quad J \nabla \mathcal{H}(x_0) = 0 \quad \Leftrightarrow \quad \nabla \mathcal{H}(x_0) = 0,$$

which shows that x_0 is a stationary point of $\mathcal{H}$. Consequently, x_0 is also an equilibrium of the system $\dot{x}(t) = f_r(x(t))$ for all $r \in \mathbb{R}$. Furthermore, we also have

$$f_r(x) = 0 \quad \Leftrightarrow \quad (J + r\mathbb{1}_{2n}) \nabla \mathcal{H}(x) = 0 \quad \Leftrightarrow \quad \nabla \mathcal{H}(x) = 0,$$

since J only has eigenvalues $\pm i$, which makes $J + r\mathbb{1}_{2n}$ nonsingular for all $r \in \mathbb{R}$. Hence, stationary points of $\mathcal{H}$ corresponds to equilibrium solutions of $\dot{x}(t) = f_r(x(t))$. Now, suppose that $x(t)$ is a solution of $\dot{x}(t) = f_r(x(t))$. Then, we have

$$\frac{d}{dt}\mathcal{H}(x(t)) = \nabla\mathcal{H}(x(t)) \cdot \dot{x}(t) = \nabla\mathcal{H}(x(t)) \cdot (J + r\mathbb{1}_{2n})\nabla\mathcal{H}(x(t)) = r\|\nabla\mathcal{H}(x(t))\|^2.$$

For $r \neq 0$, the Hamiltonian is either strictly increasing or strictly decreasing unless $x(t)$ is an equilibrium solution. For a T -periodic solution, we must have $\mathcal{H}(x(t)) = \mathcal{H}(x(t+T))$. Therefore, the system $\dot{x}(t) = f_r(x(t))$ does not admit any periodic orbit for $r \neq 0$. Using this fact, we plan to apply Theorem 4 to show that there must be periodic orbits emanating from $(r, x) = (0, x_0)$ only on the hypersurface $r = 0$. To do that, we examine the Jacobian of f_r at x_0 :

$$A(r) = \mathcal{J}_{f_r}(x_0) = (J + r\mathbb{1}_{2n})H,$$

where H is the Hessian matrix of $\mathcal{H}$ at x_0 , which is a symmetric matrix, i.e. $H^T = H$. Since $A(r)$ depends continuously on r , its eigenvalues are also continuously functions of r . Hence, there is a unique pair of complex eigenvalues $a(r) \pm ib(r)$ such that $a(0) = 0$ and $b(0) = b$ since $\pm ib$ are simple eigenvalues of $A(0)$. We claim that these eigenvalues $a(r) \pm ib(r)$ cross the imaginary axis at $\pm bi$ when $r = 0$.

First of all, $A_0 = A(0) = JH$ is nonsingular since it has no eigenvalue that is an integer multiple of bi (including $0 \cdot bi$). Due to the invertibility of J , H is also nonsingular. Furthermore, if λ is an eigenvalue of $A(r)$, then $-\bar{\lambda}$ is an eigenvalue of $A(-r)$ since $J^T = -J$ and

$$\begin{aligned} 0 &= \det[(J + r\mathbb{1}_{2n})H - \lambda\mathbb{1}_{2n}] = \det(H^{-1})\overline{\det[(J + r\mathbb{1}_{2n})H - \lambda\mathbb{1}_{2n}]^\dagger} \det(H) \\ &= \det(H^{-1}) \det[H(-J + r\mathbb{1}_{2n}) - \bar{\lambda}\mathbb{1}_{2n}] \det(H) \\ &= \det[-(J - r\mathbb{1}_{2n})H - \bar{\lambda}\mathbb{1}_{2n}] \end{aligned}$$

$$\Rightarrow 0 = (-1)^{2n} \det[(J - r\mathbb{1}_{2n})H + \bar{\lambda}\mathbb{1}_{2n}] = \det[A(-r) + \bar{\lambda}\mathbb{1}_{2n}],$$

where † denotes the operation of Hermitian adjoint, i.e. taking transpose and complex conjugation simultaneously. Therefore, $-a(r) \pm ib(r)$ are eigenvalues of $A(-r)$, which implies that $a(-r) = -a(r)$ and $b(-r) = b(r)$.

Finally, we need to prove that $a(r) \neq 0$ for $r \neq 0$. Consider the linear equation

$$\dot{x}(t) = JHx(t) = A(0)x(t),$$

which is a Hamiltonian system with corresponding Hamiltonian $\mathcal{H}_1(x) = \frac{1}{2}x^T Hx$. Indeed, we can check that $\nabla\mathcal{H}_1(x) = Hx$. Then, this system can also be embedded into a one-parameter family of differential system

$$\dot{x}(t) = (J + r\mathbb{1}_{2n})\nabla\mathcal{H}_1(x(t)) = (J + r\mathbb{1}_{2n})Hx(t) = A(r)x(t).$$

As we have shown before, the system $\dot{x}(t) = A(r)x(t)$ does not admit any periodic orbit for all $r \neq 0$. Therefore, the matrix $A(r)$ cannot have any purely imaginary eigenvalue for $r \neq 0$.

As a result, we proved the claim that the eigenvalues $a(r) \pm ib(r)$ cross the imaginary axis at $\pm bi$ when $r = 0$. By Theorem 4, every neighborhood of x_0 contains a periodic orbit when $r = 0$. $\square$

The condition that no other eigenvalue of the Jacobian at a center is an integer multiple of the purely imaginary eigenvalue is called the *nonresonance condition*. Without this condition, it can happen that the system does not have periodic solutions around the center. This shows that a solution to a nonlinear equation in a small neighborhood does not necessarily correspond to a solution to the linearized equation in this neighborhood. The following example illustrates this phenomenon [5, Example 9.2].

Example 5. Consider the Hamiltonian $\mathcal{H} : \mathbb{R}^4 \rightarrow \mathbb{R}$ given by

$$\mathcal{H}(x) = \mathcal{H}(x_1, x_2, x_3, x_4) = \frac{1}{2} (x_1^2 + x_3^2 - x_2^2 - x_4^2) + (x_1^2 + x_2^2 + x_3^2 + x_4^2)(x_3x_4 - x_1x_2).$$

First of all, we have

$$\nabla \mathcal{H}(x) = \begin{pmatrix} x_1 + 2(x_3x_4 - x_1x_2)x_1 - (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_2 \\ -x_2 + 2(x_3x_4 - x_1x_2)x_2 - (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_1 \\ x_3 + 2(x_3x_4 - x_1x_2)x_3 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_4 \\ -x_4 + 2(x_3x_4 - x_1x_2)x_4 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_3 \end{pmatrix}.$$

Then, the corresponding Hamiltonian system is

$$\dot{x} = J\nabla \mathcal{H}(x) = \begin{pmatrix} x_3 + 2(x_3x_4 - x_1x_2)x_3 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_4 \\ -x_4 + 2(x_3x_4 - x_1x_2)x_4 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_3 \\ -x_1 - 2(x_3x_4 - x_1x_2)x_1 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_2 \\ x_2 - 2(x_3x_4 - x_1x_2)x_2 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_1 \end{pmatrix}.$$

Observe that $x(t) = 0$ is an equilibrium solution of this equation. If $x(t)$ is a solution of this equation, then

$$\begin{aligned} \frac{d}{dt}(x_1x_4 + x_2x_3) &= \dot{x}_1x_4 + x_1\dot{x}_4 + \dot{x}_2x_3 + x_2\dot{x}_3 \\ &= x_3x_4 + 2(x_3x_4 - x_1x_2)x_3x_4 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_4^2 \\ &\quad + x_1x_2 - 2(x_3x_4 - x_1x_2)x_1x_2 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_1^2 \\ &\quad - x_3x_4 + 2(x_3x_4 - x_1x_2)x_3x_4 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_3^2 \\ &\quad - x_1x_2 - 2(x_3x_4 - x_1x_2)x_1x_2 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)x_2^2 \\ &= 4(x_3x_4 - x_1x_2)^2 + (x_1^2 + x_2^2 + x_3^2 + x_4^2)^2. \end{aligned}$$

Since the right-hand side is positive for $x(t) \neq 0$, the quantity $x_1(t)x_4(t) + x_2(t)x_3(t)$ is strictly increasing when the orbit leaves the origin. If $x(t)$ is a periodic solution, then the quantity $x_1(t)x_4(t) + x_2(t)x_3(t)$ must be also periodic, which contradicts with above fact. Therefore, this system has no periodic orbit.

On the other hand, the linearized equation around the origin is

$$\dot{x}(t) = \begin{pmatrix} x_3(t) \\ -x_4(t) \\ -x_1(t) \\ x_2(t) \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} x(t).$$

The characteristic equation of the above matrix is

$$0 = \det \begin{vmatrix} -\lambda & 0 & 1 & 0 \\ 0 & -\lambda & 0 & -1 \\ -1 & 0 & -\lambda & 0 \\ 0 & 1 & 0 & -\lambda \end{vmatrix} = (\lambda^2 + 1)^2 = (\lambda + i)^2(\lambda - i)^2,$$

which has two solutions $\lambda = \pm i$. Each of these eigenvalues has a multiplicity 2, which violates the nonresonance condition. This shows that the linearized equation has 2π -periodic orbits while the original differential equation does not admit any periodic orbit.

Various conditions implying the existence of periodic orbits around a center of a Hamiltonian system have been studied until now [6].

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